

SUBASHINIE M

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SUMMARY

M.Sc. Artificial Intelligence & Data Science student with hands-on experience across full-stack web development and applied machine learning. Built four independent projects spanning web applications (Flask, MySQL, Razorpay), NLP, classical ML, and fuzzy logic, alongside a healthcare AI diagnostic pipeline built during an internship. Strong in Python, Flask, SQL, and end-to-end application development, with an interest in both software engineering and data-driven roles.

EDUCATION

M.Sc. Artificial Intelligence & Data Science

2025 - 2027

Karunya Institute of Technology and Sciences, Coimbatore | CGPA: 9.3

Bachelor of Computer Applications (BCA)

2022 - 2025

Madras Christian College, Chennai | CGPA: 8.36

EXPERIENCE

Healthcare AI Intern | Margy Tech Pvt Ltd

Jun 1 - Jun 30, 2026

- Built a CT Brain Study AI Analysis System (Python, Flask) integrating a hospital PACS server via DICOMweb to retrieve and analyze real CT Brain DICOM studies
- Trained a Normal/Abnormal classifier on ~400 CT studies (200/200 split, 70/15/15 train-val-test) — pretrained ResNet18 feature extraction feeding an XGBoost classifier (1000 trees), achieving 72.5% ROC-AUC on validation
- Built a 6-step DICOM preprocessing pipeline (Hounsfield Unit conversion, windowing, normalization, resizing) and developed 4 Flask routes powering the end-to-end interface

PROJECTS

Online Venue Reservation System

Python, Flask, MySQL, Razorpay, APScheduler

- Built a full-stack booking platform with role-based access (admin/user); load-tested with 100 concurrent users and 200+ booking transactions with zero double-booking conflicts, sub-2-second response time
- Engineered an APScheduler job to auto-cancel unpaid bookings after a 2-day window, with Flask-Mail confirmations delivered within 3 seconds; integrated Razorpay for payments. Published as a paper in the BCA department journal

Road Accident Severity Prediction & Analysis

Python, Flask, Scikit-learn, Pandas, Seaborn

- Built an end-to-end ML pipeline (data cleaning → training → Flask deployment) using a 200-tree Random Forest classifier to predict 3-class accident severity (Fatal/Serious/Slight) from 6 engineered features, validated on an 80/20 train-test split
- Built a Flask analytics dashboard with 6 interactive visualizations (yearly trend, state-wise heatmap, weather correlation, severity distribution) to surface accident hotspots

NLP Chatbot for Student Query Support

Python, Flask, Scikit-learn, NLTK

- Built a TF-IDF + Logistic Regression intent classifier with domain-specific synonym expansion, covering 13 query categories (admissions, fees, hostel, exams, etc.) with sub-1-second response time
- Designed multi-intent detection (returns up to 3 ranked intents) with a cosine-similarity fallback for unmatched queries, deployed via a real-time Flask interface

Photo Shield – Fake Image Detector

Python, Flask, OpenCV, scikit-fuzzy, SciPy

- Built a fake/morphed image detector combining probability theory, entropy analysis, and Error Level Analysis (ELA) to generate a normalized 0-1 authenticity score from image features
- Designed a fuzzy inference system with rule-based logic mapping entropy/ELA to Low/Medium/High risk, producing a 0-100 Fuzzy Risk Score and 3-way classification (Original / Morphed / AI-Generated)
- Built a Flask web interface for image upload/URL analysis, displaying confidence score, fuzzy risk score, and detection metrics in real time

TECHNICAL SKILLS

Languages: Python, Java (basics), SQL

Web Development: HTML, CSS, Flask

Databases: MySQL, MongoDB

Machine Learning & Computer Vision: Pandas, NumPy, Scikit-learn, OpenCV, ResNet18, XGBoost

Tools: Git, GitHub

SOFT SKILLS

- Team player
- Curious learner
- Strong communication
- Time management

ACHIEVEMENTS

National-Level Ideathon — CIVORA: Building a Safer Urban Aura Through Edge AI

Team SparkUnion | Karunya Institute of Technology and Sciences

- Participated in a national-level ideathon as part of a 5-member team, co-developing and presenting CIVORA — an Edge AI smartphone concept for real-time road hazard detection, civic infrastructure reporting, and automatic crash-evidence capture for gig-economy drivers